

This is a recall of linear algebra, very fundamental.

Recall the base change matrix is $(e_1 \dots e_n) A = (f_1 \dots f_n)$

$$\sum x_i e_i = (e_1 \dots e_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \Leftrightarrow \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = A \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$\text{by } \sum x_i e_i = (e_1 \dots e_n) A \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = (f_1 \dots f_n) \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \sum y_i f_i$$

stay invariant $v = \sum x_i e_i = \sum y_i f_i$

$\Rightarrow V \xrightarrow{\text{id}} V$ composite isomorphisms to $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$

$$\begin{matrix} \uparrow & \uparrow \\ (e_1 \dots e_n) & (f_1 \dots f_n) \end{matrix}$$

$\mathbb{R}^n \xrightarrow{T} \mathbb{R}^n$ (canonical basis $\Leftrightarrow \mathbb{R}^n$), reduce to $\mathbb{R}^n$ to write as matrix

Due to the representation matrix of operator T under basis $(e_1 \dots e_n)$ in source and $(f_1 \dots f_n)$ in target is $\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

$$\Rightarrow T \rightsquigarrow A^T$$

That is: Base change matrix from basis $(e_1 \dots e_n)$ to basis $(f_1 \dots f_n)$

= Representation matrix of id under basis $(f_1 \dots f_n)$ to $(e_1 \dots e_n)$

= Representation matrix of $\tilde{T}: \sum x_i e_i \mapsto \sum x_i f_i$ under basis $(e_1 \dots e_n)$

This is by $\sum x_i f_i = (f_1 \dots f_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (e_1 \dots e_n) \boxed{A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}}$, or by diagram,

we justify the three isomorphism in the square, s.t. $\mathbb{R}^n \xrightarrow{\text{id}} \mathbb{R}^n$

$$\begin{array}{ccc} V & \xrightarrow{\tilde{T}} & V \\ \uparrow & & \uparrow \\ (e_1 \dots e_n) & & (f_1 \dots f_n) \\ \mathbb{R}^n \xrightarrow{\text{id}} \mathbb{R}^n \end{array}$$

Although isomorphism (noncanonical) doesn't matter in most cases, justify isomorphism to another isomorphism to create id strictly is important trick in higher theory, e.g. deformation theory with derivations ...

Exercise. Determine the operator $\tilde{T}$, s.t. its representation matrix is A above, under basis $(f_1 \dots f_n)$

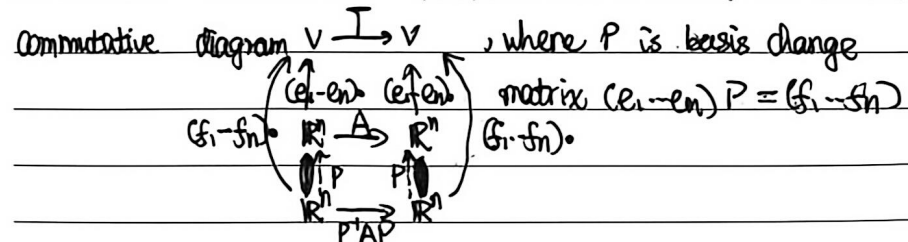


Thus ~~we~~ we have category equivalence between
 {Category of Vector Spaces, with fixed basis} $\leftrightarrow$ $\{ \mathbb{R}^0, \mathbb{R}^1, \mathbb{R}^2, \dots \}$
 non-degenerate linear morphism $\leftrightarrow$ {Matrix}

When we consider ~~a~~ morphism between same vector space with different basis, it's related with base change matrix.

About a half of ~~time~~ time when we first studied linear algebra was used to ~~the~~ compute the Ker/Coker, Dual/Hom and Tensor and their compatibility via the right side matrix model.

Return to the base change, now consider endomorphism, i.e. $V \rightarrow V$ with same basis, then we change both source and target to a same basis $\Rightarrow A \rightsquigarrow P^T A P$ in matrix model as we have



This means that A and $P^T A P$ induces same linear transform, i.e. {Linear transform} $\leftrightarrow$ {Matrix} $\{ A \sim P^T A P \mid P \in GL_n(\mathbb{R}) \}$

There are many equivalence relation in matrices, the next is by reduce $GL_n(\mathbb{R})$ to some matrix subgroup, for example, $\{ A \sim P^T A P \mid P \in O(n) \}$ is used in the study of quadratic forms, and it allows us to define projection operators;

Another, and the strongest is $\{ A \sim P A Q \mid P, Q \in GL_n(\mathbb{R}) \}$
 $\Rightarrow A \sim \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ as $P, Q \leftrightarrow$ operation on matrix A

it has the advantage that, it preserves ~~the~~ useful invariants: ~~det/rank/...~~ ^{most} ~~det/rank/...~~, which is a strong consequence of vector space as free module



Q1. Can we use $\{ A \sim P^T A P \}$, reduce all A to $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$?

No, $A \sim P^T A P$ preserves $\det$, but $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ has $\det = 0$ or ± 1

Q2. Can we use $\{ A \sim P^T A P \}$, reduce all A to $\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$?

This is not so obvious, but you can easily find a counterexample, after learning linear algebra, we know diagonalizable matrix is not so general, it means semi-simple.

~~Exercise~~ Find a matrix not semi-simple

To be closer to Q2, we interpret it as two versions:

Q2'. Can we justify basis of V s.t. they're eigen-vector of T , then $A \leftrightarrow \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$ i.e. $V = \bigoplus V_{\lambda_i}$, $T \mapsto V_{\lambda_i} \mapsto \begin{pmatrix} \lambda_i & & \\ & \ddots & \\ & & \lambda_i \end{pmatrix}$?

Q2". We have ${}^{\lambda} \mathbb{R}[T] \hookrightarrow V$, i.e. V as a $\mathbb{R}$ -module is extended to a $\mathbb{R}[T]$ -module, can we make $P^T A P$ simpler via decompose V ?

These two forms lead to two normal forms, which are totally different

A2'. It's all most diagonalizable, i.e. the Jordan-Chevalley decomposition

$A \sim \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} + N$, N nilpotent part, ~~this is called~~ $\begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix}$ Jordan normal form

This holds for general Lie group/algebra-rep, we can find the maximal torus $\subset$ Lie group/algebra, then get semi-simple rep operator for the maximal torus, last fill other operators.

A2". By the structure thm of f.g. module over PID, applied to $\mathbb{R}[T] \hookrightarrow V \Rightarrow V \cong \mathbb{R}[T]/(f_1) \oplus \dots \oplus \mathbb{R}[T]/(f_k)$, $f_1 \dots f_k = \det(\lambda I - T)$

s.t. $T \mapsto \mathbb{R}[T]/(f_1) \oplus \dots \oplus \mathbb{R}[T]/(f_k)$ ^{minimal polynomial of T}
 has matrix form $T \mapsto \mathbb{R}[T]/(f_j)$ as $\begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix}$

this is called rational normal form. ^{Coefficient of f_j/f_{j-1}}



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* Side story: $O(n)$, theory of quadratic form, ~~and~~ conic, and projective compactification in geometry.

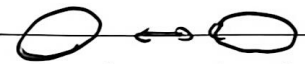
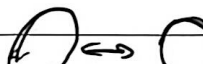
Recall the quadratic form $x^T A x = 0$ is always homogenous ~~and~~, to correspond to conic $\subset \mathbb{R}^2$, we extend it by $\begin{pmatrix} x \\ 1 \end{pmatrix}^T \begin{pmatrix} A & b \\ b^T & c \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = 0$ where $A \in M_2(\mathbb{R})$ can degenerate, and we can make it symmetric. Thus $\{\text{planar conic}\} \leftrightarrow \{A \in M_2(\mathbb{R}), b \in \mathbb{R}^2, c \in \mathbb{R} \text{ triple} \mid A = A^T\}$ now we put equivalence relations on left, as it's the most fundamental problem in geometry.

• Isometry $\leftrightarrow \tilde{A} \sim P^T \tilde{A} P$ for $P \in O(3)$

• Affine $\leftrightarrow \tilde{A} \sim P^T \tilde{A} P$ for $P \in GL_3(\mathbb{R})$

$\Rightarrow \tilde{A} \sim \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ or $\begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$

isometry preserves eigenvalue, but affine transform only preserve

 $\leftrightarrow$  their sign. There former classify by sign ~~and~~ the latter classification.

The result is (even the proof is easy):

$\begin{cases} a_1 a_2 > 0 \text{ elliptic} \\ a_1 a_2 = 0 \text{ parabolic} \\ a_1 a_2 < 0 \text{ hyperbolic} \end{cases}$

Rk. These products

can be directly computed via determinant, without change to normal form

Problem: ① Imaginary ones

② Degenerate case $a_1 a_2 \rightarrow 0, b \rightarrow 0, c \rightarrow 0 \dots$

Solution: ① Replace $\mathbb{R}$ by algebraically closed field

② Homogenization/Projectization

This is the origin / first example of the classification problem,

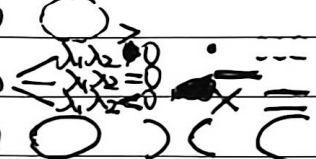
in algebraic geometry.

Namely, we take projectization:

$GL_3(\mathbb{R}) \hookrightarrow P\left(\begin{pmatrix} A & b \\ b^T & c \end{pmatrix} \in M_3(\mathbb{R})\right) \leadsto PGL_3(\mathbb{R}) \hookrightarrow PGL_3(\mathbb{R})$

$$\begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \leadsto \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

The projective equivalence classes are easier by $\lambda_1, \lambda_2, \lambda_3$

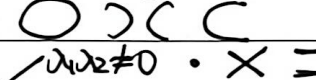
$\begin{cases} \lambda_1 \lambda_2 \lambda_3 > 0 \\ \lambda_1 \lambda_2 \lambda_3 = 0 \\ \lambda_1 \lambda_2 \lambda_3 < 0 \end{cases}$  $\begin{cases} x_1^2 + x_2^2 + x_3^2 = 0 \\ x_1^2 + x_2^2 = 0 \\ x_1^2 - x_2^2 = 0 \\ x_1^2 + x_2^2 - x_3^2 = 0 \end{cases}$ } Only 5 classes

for example, $x_1^2 + x_2^2 - x_3^2 = 0 \rightarrow x_1 = 1$ hyperbolic } choosing different affine open set
 $\rightarrow x_3 = 1$ elliptic
 $\rightarrow x_2 = 1$ parabolic

Exercise: Choose affine open

sets, s.t. $x_1^2 + x_2^2 = 0$ and $x_1^2 - x_2^2 = 0$ give the four affine classes defined above

Then take $\mathbb{C}$ extending $\mathbb{R}$, consider the equations:

$\begin{cases} \lambda_1 \lambda_2 \lambda_3 \neq 0 \\ \lambda_1 \lambda_2 \lambda_3 = 0 \end{cases}$  } only 3 classes

Rk. This works for all ~~quadratic~~ quadratic hypersurfaces ~~for~~ for $X \subset \mathbb{A}^n$,

we consider $GL_{n+1}(\mathbb{C}) \hookrightarrow P\left(\begin{pmatrix} A & b \\ b^T & c \end{pmatrix} \in M_{n+1}(\mathbb{C})\right)$

or projectization $PGL_{n+1}(\mathbb{C}) \hookrightarrow PGL_{n+1}(\mathbb{C})$, but for $\deg(X)$ higher,

it becomes more difficult: to parameterize the coefficients of higher

degree polynomial, ~~to~~ we need a "cube" then matrix, thus it's

nontrivial to build such a theory as "higher matrix", see

[Discriminants, Resultants and Multidimensional Determinants]